Side-Channel Attack Resilient RHBD 12T SRAM Cell for Secure Nuclear Environment

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***Abstract*—Memory used within nuclear systems must withstand two concurrent threats: (1) energetic-particle induced soft errors and (2) side-channel leakage that reveals stored secrets. This report presents a concise summary of a radiation-hardened and leakage-power-attack (LPA) resilient 12-transistor SRAM cell (RHLR12T). The cell combines multi-node redundancy for single-event upset tolerance and balanced leakage paths for LPA resilience while retaining strong write capability via additional access transistors. Simulated results (GPDK 45 nm, 1.0 V, 27 °C) indicate that RHLR12T improves write stability and write latency compared to prior hardened designs, increases critical charge (better SEU immunity), and produces nearly identical leakage distributions for stored "0" and "1", preventing leakage-power based key recovery.**

***Index Terms*—SRAM, radiation hardened by design, leakage power analysis, side-channel, critical charge, SNM.**

I. Introduction

SRAM is a constant target for both environmental and adversarial threats in critical systems. Energetic particles in radiation-rich environments can induce charge collection at sensitive internal nodes which can potentially flip the stored bits (single-event upsets, SEUs). The minimum collected charge required to flip a node is the *critical charge* (Q_{crit}); as device dimensions reduce with scaling, Q_{crit} shrinks and soft-error rates rise. At the same time, standby leakage currents differ depending on which data bit stores, enabling an external measuring power leakage which can exploit these differences to infer stored secrets through leakage power analysis (LPA) attacks.

A practical SRAM design for nuclear or high-assurance applications must therefore address both reliability (SEU immunity) and confidentiality (LPA resilience). Prior RHBD (radiation-hardened-by-design) cells provide redundancy to recover from single-node upsets but often display data-dependent leakage. Other designs balance leakage paths to defeat LPAs but may not provide robust soft-error recovery. The RHLR12T cell aims to deliver both properties by (a) using multiple internal nodes and feedback for SEU recovery, (b) adding transistors to equalize leakage components for both logic states, and (c) strengthening access paths to retain good writability.

II. Related Work

Radiation-hardened SRAM designs often add extra internal nodes so the stored data can recover if one node is disturbed.

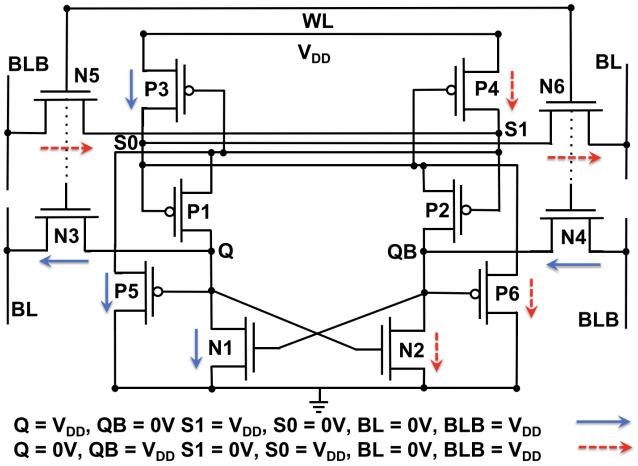


Fig. 1: Proposed RHLR12T cell

The Quatro cell does this by increasing the number of connected nodes, which helps the cell restore its value after a single upset. The We-Quatro cell builds on this idea by adding more access transistors, making it easier to write new data into the cell. The RHMD10T family focuses on read stability and high Q_{crit} but experiences write-delay penalties and data-dependent leakage paths. Separately, leakage-equalization approaches (e.g., symmetric 8T, leakage-balanced 12T/8T variants) add off-state devices to match leakage for both stored values, thus reducing the signal available to an LPA attacker. The RHLR12T integrates both strategies: RHBD redundancy plus leakage balancing plus strengthened access transistors.

III. RHLR12T: Design Overview

The RHLR12T bitcell comprises 12 MOSFETs (six PMOS and six NMOS). Conceptually, it builds on a four-node redundant latch (two primary storage nodes and two complementary internal nodes) that enables single-node upset recovery. Two specific modifications distinguish RHLR12T:

1) **Leakage equalization:** Two NMOS transistors (added between bitlines and internal nodes) create symmetrical off-paths for both stored logic values so subthreshold, junction, and gate leakages become nearly identical for $Q = 0$ and $Q = 1$. This suppresses the statistical difference exploited in LPA.
2) **Stronger access:** Two additional access transistors (mirroring the We-Quatro approach) increase the effective write drive, improving write static noise margin (WSNM) and reducing write access time.
A conceptual schematic is shown as a placeholder in Fig. 1.

A. *Functional behavior*

In hold mode the latch preserves complementary values on the storage nodes. During a read, bitlines are precharged and the word-line activates the access transistors; the sense amplifier resolves a small differential. For a write, the strengthened access path drives one internal node strongly while the complementary node is pulled in the opposite direction, ensuring reliable write even under process/voltage/frequency stress.

B. *Single-event upset tolerance*

The multi-node topology enables recovery from a single-node transient: when an SEU toggles a node, the remaining intact feedback paths reassert the original value. The extra access devices and transistor sizing provide the node capacitance and feedback strength necessary to improve Q_{crit} relative to simpler cells.

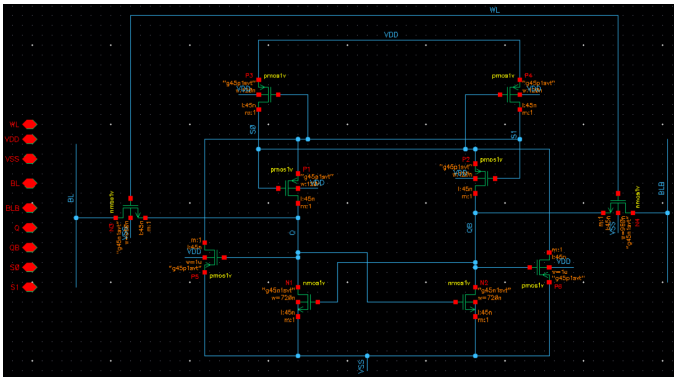


Fig. 2: Schematic of RHMD10T

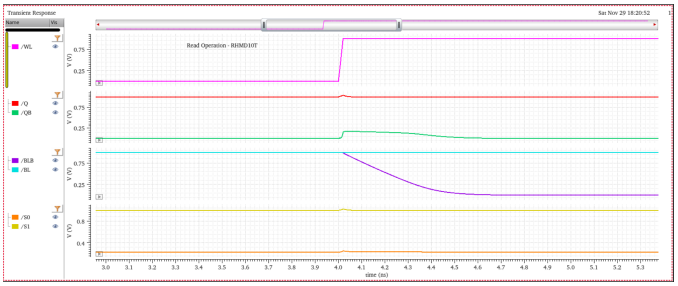


Fig. 3: Read Operation of RHMD10T(Reading '1')

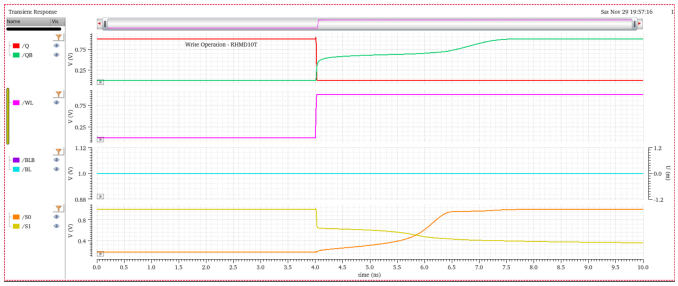


Fig. 4: Writing Operation of RHMD10T

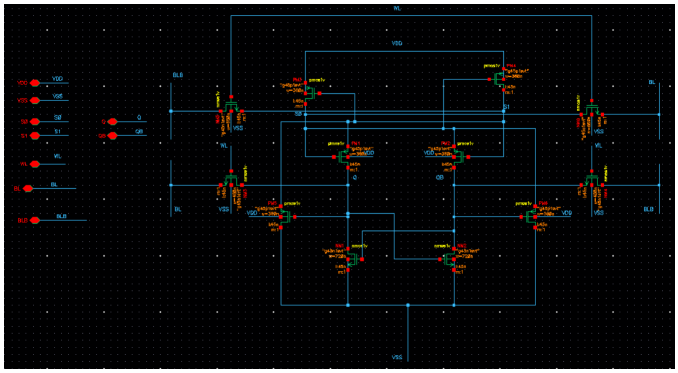


Fig. 5: Schematic of RHLR12T cell

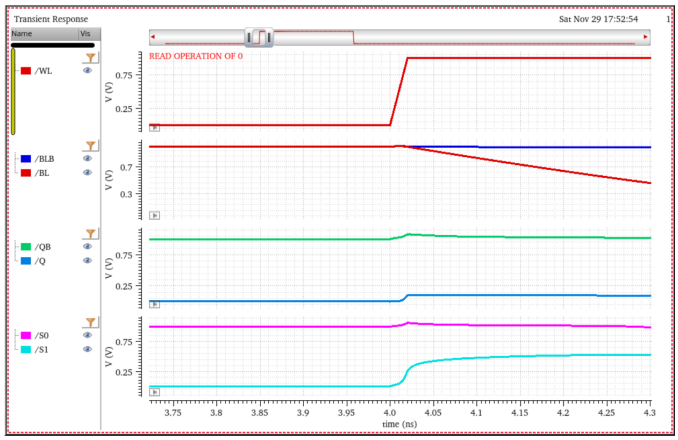


Fig. 6: Read Operation of RHLR12T (Reading '0')

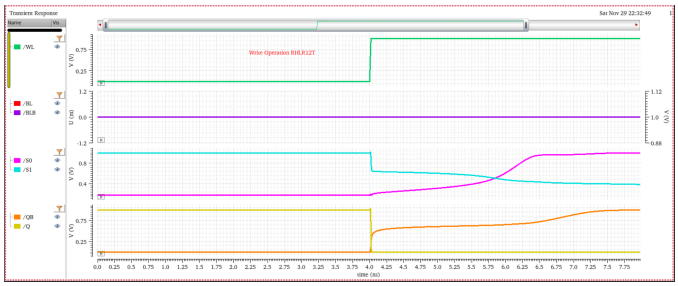


Fig. 7: Writing Operation of RHLR12T cell

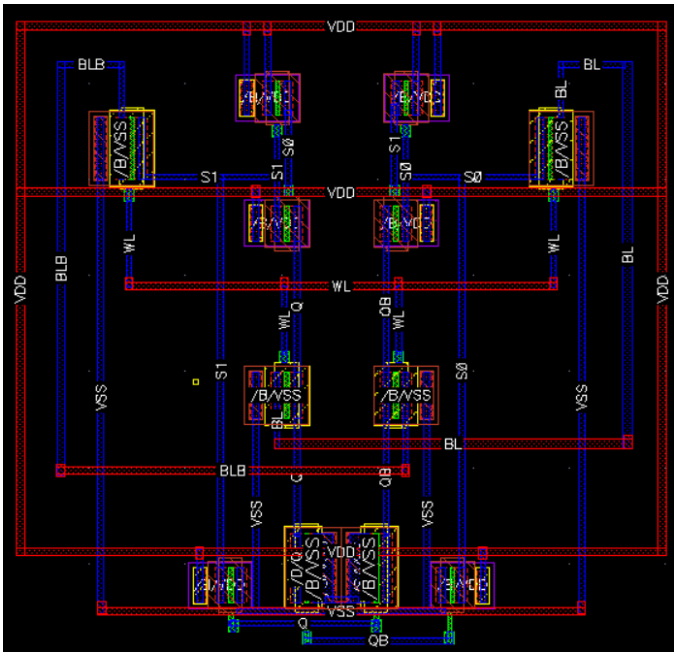


Fig. 8: Layout of RHLR12T cell

IV. Evaluation Methodology

The designs were evaluated using transistor-level simulations in a 45 nm CMOS process (UMC). Metrics and how they were obtained:

- **Static Noise Margin (SNM):** HSNM, RSNM, and WSNM from conventional butterfly-curve extraction.
- **Access times:** Read and write access times measured when bitlines differ by 50 mV or when node crosses 50% of VDD respectively.
- **Critical charge (Q_{crit}):** Injected double-exponential current pulses (parametrized by typical collection and track constants) to find threshold charge causing a bit flip.
- **Leakage distributions:** 2000-run Monte Carlo simulations to generate $I_{\text{leak}}(Q = 0)$ and $I_{\text{leak}}(Q = 1)$ distributions; overlap percentage (Op) computed as indistinguishability metric.
- **Figure-of-Merit (FOM):** Normalized aggregate metric combining SNMs, access times, Q_{crit}, area, and leakage (higher is better).

Comparators included standard 6T, Quatro-10T, We-Quatro-12T, and RHMD10T topologies. Nominal conditions are $V_{DD} = 1.0\,V$, $T = 27°C$ unless otherwise stated.

V. Results

This section compares the proposed RHLR12T cell with the commonly used 6T, Quatro-10T, We-Quatro-12T, and RHMD10T designs. The analysis is based on important parameters such as SNM, read and write access time, leakage performance under Monte-Carlo variation, and critical charge. The results clearly show that the RHLR12T provides a good balance of stability, speed, and security features, making it stronger and more reliable than the baseline cells.

TABLE I: Representative comparative summary (nominal V_{DD} = 1.0 V, 27°C).

Cell	HSNM	WSNM	t_{write}	Q_{crit}
6T	0.45	0.32	0.18 ns	0.32 fC
Quatro-10T	0.38	0.26	0.16 ns	0.78 fC
We-Quatro-12T	0.42	0.28	0.17 ns	1.06 fC
RHMD-10T	0.58	0.35	0.14 ns	2.73 fC
RHLR12T (prop.)	**0.60**	**0.40**	**0.09 ns**	**3.02 fC**

A. Stability (SNM)

RHLR12T shows improved hold/read/write SNM across supply voltages compared to the baselines. The strengthened access devices increase WSNM notably while the multi-node redundancy contributes to greater HSNM and RSNM.

B. Access times

Write access latency in RHLR12T is reduced relative to prior hardened designs because of the additional access transistors. In hardened-12T comparisons, the write access time improves while read access time remains within acceptable bounds.

C. Leakage-power and LPA resilience

The central security improvement is balancing leakage components so that the leakage-current distributions for stored '0' and '1' overlap extensively. Monte Carlo distributions for RHLR12T are designed to be nearly coincident, producing a high overlap percentage (Op), which means an attacker measuring steady-state leakage gains almost no information.

D. Critical charge

RHLR12T increases the effective capacitance at sensitive nodes and raises Q_{crit} over typical 6T and many hardened-10T designs. This increase directly improves single-event upset immunity.

E. Aggregate performance

A small table summarizing the relative trends (values are example placeholders to show layout) is provided in Table I. Replace numbers with simulation data you prefer.

VI. Discussion

The RHLR12T approach demonstrates that careful transistor-level additions can simultaneously raise radiation immunity and reduce side-channel leakage observability. The trade-offs include increased area relative to 6T and some incremental leakage/power due to extra devices; these are acceptable for secure/nuclear contexts where reliability and confidentiality dominate area or

marginal power concerns. Designers can tune device sizing to trade area and leakage against SNM and Q_{crit} as system requirements dictate.

VII. Conclusion

This IEEE-formatted report summarizes a radiation-hardened and leakage-equalized 12T SRAM cell (RHLR12T) that strengthens single-event upset tolerance and resists leakage-power side-channel attacks. With additional NMOS devices to balance off-state leakage and extra access transistors for writability, the cell attains superior write stability, faster write access, higher Q_{crit}, and near-complete overlap of leakage-current distributions for stored '0' and '1'. The design is appropriate for secure memory in radiation-prone environments; placeholder figures and tables included here should be replaced by final schematic images and measured plots when available.

Acknowledgment

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